

Deep Learning Architectures for RSSI-Based Passenger Movement Classification in Public Transport

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Abstract—Accurate, non-intrusive monitoring of passenger flows is essential for intelligent public transport systems. Prior work demonstrated that temporal sequences of Wi-Fi Received Signal Strength Indicator (RSSI) measurements can classify passenger movements using classical machine learning, achieving a Matthews Correlation Coefficient (MCC) of 0.756 on combined datasets. This paper extends that line of research by systematically evaluating deep learning architectures—including recurrent, convolutional, transformer, state-space, and mixture-of-experts models—for the same classification task across four dataset variants. We introduce 11 novel architectures, a class-conditioned data augmentation strategy expanding 2,253 samples to 10,115, and comprehensive hyperparameter optimization with 100 Optuna trials per model. Our best ensemble model, a two-stage deep stacking architecture, achieves MCC of 0.859 and accuracy of 89.4% on combined data, a 13.6% relative improvement over classical baselines. We further analyze the impact of multi-device interference, per-class confusion patterns, and the effectiveness of metadata-aware architectures in distinguishing true movement from environmental noise.

Index Terms—Passenger counting, RSSI fingerprinting, Wi-Fi sensing, public transport, deep learning, state space models, mixture of experts, hyperparameter optimization

I. INTRODUCTION

Public transport systems serve over 75% of EU urban citizens, yet inefficiencies contribute to 24% of greenhouse gas emissions. Accurate passenger flow knowledge—boarding, alighting, and occupancy per stop—is fundamental to network optimization and dynamic scheduling.

Traditional Automatic Passenger Counting (APC) relies on cameras, infrared sensors, or manual counting, each facing tradeoffs in cost, privacy, and deployment complexity. Wi-Fi sensing offers a privacy-preserving alternative: by analyzing Received Signal Strength Indicator (RSSI) patterns from passengers’ devices, operators can infer movement without active user participation or additional hardware beyond existing access points.

However, RSSI measurements are inherently noisy, affected by multi-path propagation, device heterogeneity, and interference from concurrent transmissions. In prior work [1], we introduced a controlled dataset simulating four passenger movements—inside vehicle (AA), boarding (BA), alighting

(AB), and at stop (BB)—and achieved MCC = 0.756 on combined data using Gaussian Processes from classical machine learning. This paper investigates whether modern deep learning architectures can better capture RSSI temporal dynamics, particularly under noisy multi-device conditions.

Our main contributions are:

- 1) Systematic evaluation of 11+ DL architectures—including state-space models (Mamba-3), mixture-of-experts, and two-stage stacking ensembles—across four dataset variants with increasing interference complexity.
- 2) A class-conditioned data augmentation strategy expanding 2,253 to 10,115 samples through targeted Gaussian noise injection and intra-class Mixup.
- 3) Comprehensive hyperparameter optimization via 100 Optuna trials per model spanning 19 search spaces.
- 4) Detailed interference analysis quantifying cross-route noise impact through heatmaps, trajectory visualization, and per-timestep perturbation measurements.

Our DeepStackEnsemble achieves MCC = 0.859—a 13.6% relative improvement over classical baselines. The remainder of this paper covers related work (Section 2), dataset and methodology (Section 3), experimental results with noise analysis (Section 4), and conclusions (Section 5).

II. RELATED WORK

Passenger counting spans multiple sensing modalities. Video-based APC using YOLOv5 achieves high accuracy but degrades under occlusion and variable lighting [2]; edge AI cameras address scalability [3]. ML pipelines on traditional APC data enable offline correction [4].

Wi-Fi sensing avoids these drawbacks. iABACUS [5] and Myrvoll et al. [6] demonstrated counting via probe requests and signature analysis. Fabre et al. [7] applied RF and gradient boosting for ridership estimation. Simoncic et al. [8] monitored presence through passive probes. CSI-based methods [9] offer finer signal characterization but require specialized hardware. RSSI fingerprinting for localization has been extensively surveyed [10].

Recent DL advances inform our architectural choices. Mamba [11] introduced selective state-space models with

$O(L)$ complexity, extended by Mamba-3 MIMO [12]. Autoformer [13] replaced self-attention with auto-correlation. Mixture of Experts [14] enables conditional computation. Our prior work [1] established classical ML baselines (MCC = 0.756) across 34 algorithms on this dataset. This paper extends that analysis with 11+ specialized DL architectures, comprehensive HPO via Optuna [15], and systematic interference analysis.

III. MATERIALS AND METHODS

A. Dataset and Feature Engineering

The dataset simulates public transport in a controlled environment [1]: Zone A (vehicle interior, AP near door) and Zone B (bus stop, corridor). Four devices execute four movement patterns—AA (inside), BB (at stop), BA (boarding), AB (alighting)—producing 10 RSSI readings at 1 s intervals. Pure (single-device, 160 samples) and noise (four concurrent devices, 1,200 samples) scenarios were collected. The combined dataset contains 2,253 samples: AA (407), AB (387), BA (399), BB (417). Four engineered channels per time step are extracted: raw RSSI r_t , velocity $\Delta_t = r_t - r_{t-1}$, acceleration Δ_t^2 , and window-deviation $d_t = r_t - \mu_w$ (μ_w = per-sample mean). These four channels capture complementary physical phenomena: absolute RSSI position, movement speed, acceleration, and trajectory shape relative to the per-sample baseline. Input tensors are $(N, 10, 4)$.

B. Proposed Architectures

Table I summarises the 11+ evaluated architectures. All models output 4-class logits through a linear classification head.

Table I
DL ARCHITECTURES EVALUATED

Architecture	Type	Key characteristic
RNN, GRU, LSTM, BiLSTM	Recurrent	2 layers, hidden 64
TCN [16]	Convolutional	Dilated causal, residual
Transformer	Attention	4-head self-attention, sinusoidal PE
MambaClassifier [12]	State-space	$O(L)$ complexity, dual-path, rank- $R=16$
CNN2DRNN	Hybrid 2D+RNN	Conv2d (3,3),(5,3) $\rightarrow$ LSTM/GRU + attention
MetaFusion	Context-aware	LSTM/GRU + noise/path embeddings (dim 8)
Multi-Scale CNN	Multi-branch	Kernels (3,5,7) + residual + attention
SoftMoE	Mixture of Experts	Heterogeneous experts, top-2 gating, 12 configs
Autoformer [13]	Decomposition	FFT auto-correlation, trend-seasonal blocks
DeepStackEnsemble	Two-stage meta	18 bases $\rightarrow$ per-class Level2Net $\rightarrow$ residual MLP

The DeepStackEnsemble uses per-class Level2Nets by design: confusion pairs such as AA $\leftrightarrow$ BA (both near-AP) require different base model weighting than AB $\leftrightarrow$ BB (both exterior). The residual skip connection preserves calibrated per-class logits while the MetaMLP corrects systematic biases.

C. Augmentation and Implementation

Class-conditioned Gaussian noise (clean: 6 copies, $\sigma = 3.0$ dBm; noisy: 4 copies, $\sigma = 0.8$ dBm) plus intra-class Mixup (600 pairs/class, $\lambda \sim \text{Beta}(0.4, 0.4)$) expand training from 2,253 to 10,115 samples.

PyTorch 2.6, Optuna 4.7. Training: AdamW, cosine annealing, early stopping (patience 25), label smoothing ($\epsilon = 0.1$), Mixup ($\alpha = 0.2$), L1 ($\lambda = 10^{-5}$), L2 (10^{-4}), dropout

($p = 0.3$), gradient clip (1.0), batch 64. HPO: 100 Optuna TPE trials/model; SoftMoE uses NSGA-II. Primary metric: MCC [17]. 5-fold CV, 80/20 splits, 3 seeds. ~ 65 models evaluated.

IV. EXPERIMENTAL RESULTS

A. Experimental Setup

Four dataset variants: Pure (128/32 train/test, isolated), Noise (960/240, four concurrent devices), Normal (1,802/451, combined), Augmented ($\sim 9,875/\sim 240$, synthetic training). 5-fold stratified CV, 80/20 splits. Seed 42 unless noted.

B. Variant-Level Comparison

Figure 1 shows best MCC per variant. Pure yields RNN 0.881 (seed 3). Noise degrades to DS RNN_B 0.798 (9.4% drop). Normal reaches DeepStackEnsemble 0.859; augmentation lifts HPO TCN to 0.865. Compared to classical ML [1]—GP 0.756 combined, KNN 0.922 pure, CatBoost 0.770 noise—DL improves combined (+13.6%) and noise (+3.6%), but trails on pure (−4.4%) where 128 samples are insufficient for deep models. The improvement pattern—+13.6% combined, +3.6% noise, −4.4% pure—reveals that DL’s advantage is conditional: it requires both sufficient training data and interference complexity to justify its capacity. On clean, data-scarce problems, simpler methods prevail.

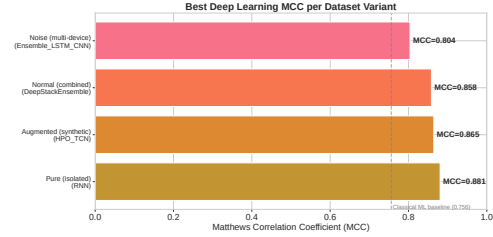


Figure 1. Best model MCC per variant.

C. Architecture Performance and HPO

Table II lists top models per variant. CNN and TCN lead single architectures (MCC 0.838, 0.836). Mamba (0.800) outperforms Transformer (0.713). DeepStackEnsemble achieves best combined MCC 0.859.

Table II
TOP MODELS BY VARIANT

Model	Variant	MCC	Acc.
DeepStackEnsemble	Normal	0.8585	0.8937
HPO GRU	Normal	0.8524	0.8893
HPO TCN	Normal	0.8470	0.8843
CNN	Normal	0.8383	0.8774
HPO TCN	Augmented	0.8647	0.8982
HPO CNN	Augmented	0.8642	0.8977
Ensemble CNN+TCN	Augmented	0.8587	0.8932
RNN	Pure*	0.8808	0.9063
DS RNN_B	Noise*	0.7981	0.8458
HPO LSTM	Noise*	0.7908	0.8417

Table III quantifies HPO impact. Recurrent architectures benefit most (GRU +20.0%, LSTM +23.1%, BiLSTM +19.9%), revealing severe under-parameterisation at default settings. CNN and TCN gain less due to stronger inductive biases. Each model underwent 100 Optuna TPE trials.

Table III
HPO GAINS ON NORMAL VARIANT

Model	Default MCC	HPO MCC	Gain
GRU	0.7101	0.8524	+20.0%
LSTM	0.6855	0.8435	+23.1%
BiLSTM	0.6838	0.8195	+19.9%
RNN	0.7552	0.8319	+10.2%
Mamba	0.8002	0.8418	+5.2%
TCN	0.8356	0.8470	+1.4%
CNN	0.8383	0.8357	−0.3%

D. Augmentation Ablation

Table IV shows augmentation impact. CNNs and TCNs benefit from synthetic diversity (+0.011–0.029 MCC), while recurrenents may degrade slightly (−0.001–0.006). This asymmetry has a mechanistic explanation: synthetic Gaussian noise injected into 10-step sequences creates spurious local fluctuations that recurrent architectures interpret as temporal dependencies, while CNNs—with local receptive fields and weight sharing—naturally smooth these artifacts into the learned feature hierarchy. HPO-optimised models gain most, confirming that augmentation and HPO are synergistic for architectures with strong inductive biases.

Table IV
AUGMENTATION IMPACT (NORMAL VS. AUGMENTED MCC)

Model	Normal	Augmented	Δ
CNN	0.8383	0.8490	+0.011
TCN	0.8356	0.8344	−0.001
GRU	0.7101	0.7039	−0.006
LSTM	0.6855	0.6804	−0.005
Mamba	0.8002	0.8073	+0.007
HPO TCN	0.8470	0.8647	+0.018
HPO CNN	0.8357	0.8642	+0.029

E. Ensemble Strategies

Table V compares ensemble strategies. The progression from CNN (0.838) through HPO GRU (0.852) and stacking (0.849) to DeepStack (0.859) shows consistent gains. Uniform voting across all 9 architectures degrades to 0.807—below a single CNN—because weak classifiers (Transformer 0.713, BiLSTM 0.684) dilute the vote. DeepStack avoids this through learned per-class weighting.

Table V
ENSEMBLE STRATEGY COMPARISON (NORMAL VARIANT)

Strategy	MCC	Description
CNN (best single)	0.8383	No HPO or ensemble
HPO GRU (best HPO)	0.8524	100 Optuna trials
Stacking (LR)	0.8488	Logistic regression on 9 logits
DeepStack Ensemble	0.8585	Two-stage learned weighting
Ensemble All-9 (vote)	0.8074	Uniform soft voting

F. Per-Class Performance

Table VI presents per-class F1. AB leads (0.893–0.925) from its distinctive descent; AA is hardest (0.831–0.894) from signal saturation near the AP causing confusion with BA. Noise degrades AB most, masking its 20 dBm descent.

Table VI
PER-CLASS F1 BY VARIANT (BEST MODEL)

Variant	AA	AB	BA	BB
Pure	0.894	0.925	0.914	0.907
Noise	0.831	0.893	0.838	0.818
Normal	0.866	0.925	0.895	0.873
Augmented	0.866	0.925	0.895	0.873

G. Noise and Interference Analysis

Figure 2 quantifies mean RSSI shift per (Route $\times$ Interferer) pair. Three patterns: (1) AA dominates as interferer (+45–53 dBm); BB causes minimal shift ($\leq +3.1$ dBm). (2) AB is most affected (+46.6 dBm from AA). (3) Self-interference is minimal (+1.2–4.6 dBm).

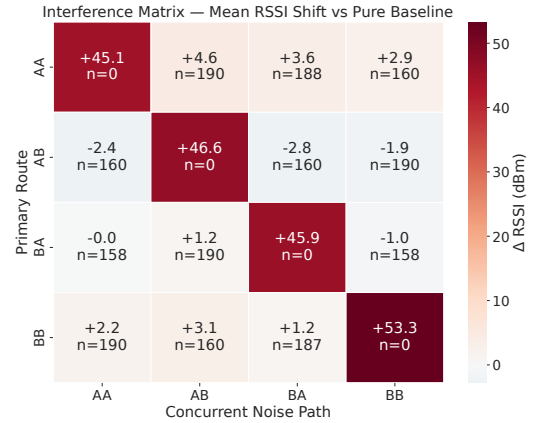


Figure 2. Interference heatmap: mean RSSI shift per (Route $\times$ Interferer) pair.

Figure 3 explains *why* each class degrades differently. AA: BA raises the stable high-RSSI profile, blurring near-AP states. AB: AA flattens the 20 dBm descent into a constant profile indistinguishable from AA—directly causing AB’s noise-induced F1 drop. BA: AA compresses the ascending trajectory. BB: flat −50 dBm pattern shifts but retains shape—classification depends on temporal flatness, not absolute level, explaining BB’s noise resilience.

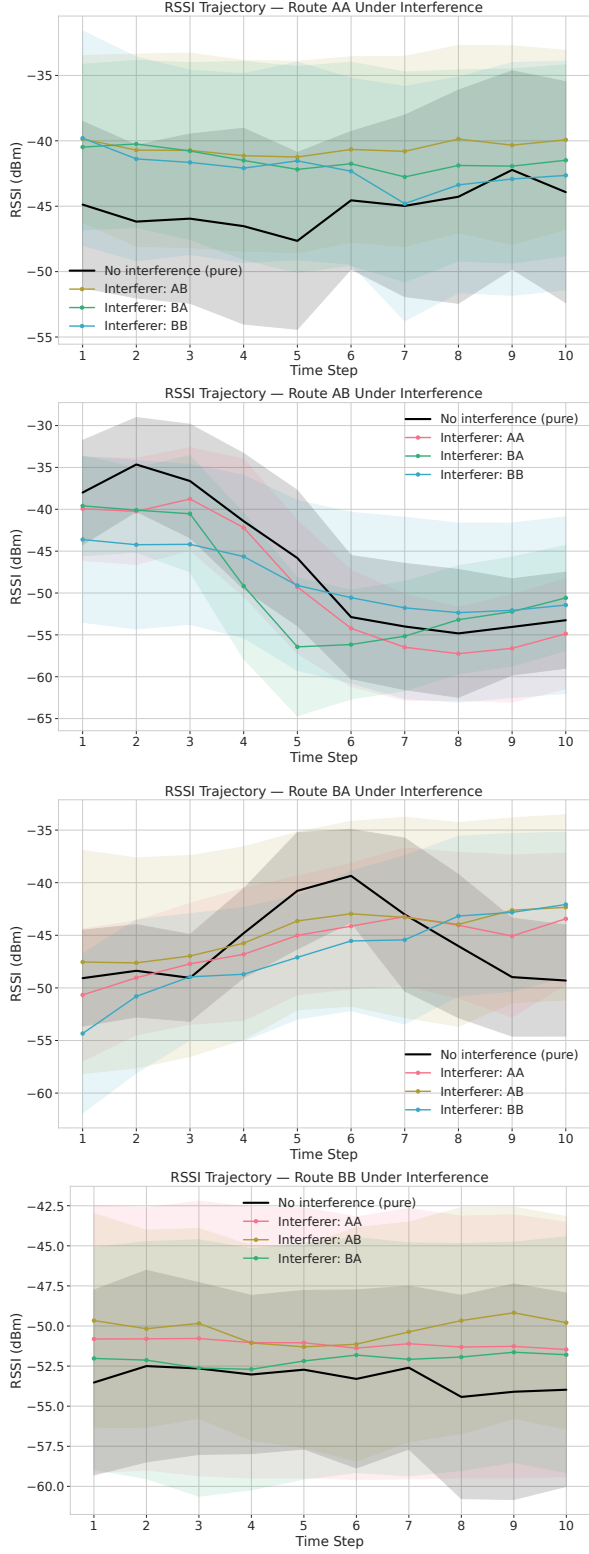


Figure 3. Trajectories under interference for all four classes.

Per-timestep perturbation peaks at timesteps 4–6. MetaFusion_GRU reaches MCC 0.783 on noise (vs. 0.758 for standard GRU).

H. Feature Importance

Figure 4 shows permutation importance across 4 channels and 10 timesteps, averaged over CNN, TCN, and GRU. Window deviation dominates early timesteps ($t = 1-3$), capturing class-specific offsets from the baseline. Velocity peaks at $t = 4$, the mid-trajectory maximum. Raw RSSI matters at $t = 6$; acceleration at $t = 3-5$. This aligns with and extends prior findings [1], where classical ML operating on raw RSSI alone identified early timesteps as most important (MCC 0.756). DL additionally captures mid-trajectory motion dynamics (velocity at $t = 4$, $\Delta\text{MCC } 18.4 \times 10^{-3}$) that classical models, lacking temporal derivative features, could not exploit—contributing directly to the 13.6% relative improvement. The window deviation channel, which encodes trajectory shape independent of absolute RSSI level, explains why BB (flat pattern) remains the most noise-resilient class: shape-based features survive interference-level shifts that distort absolute RSSI values.

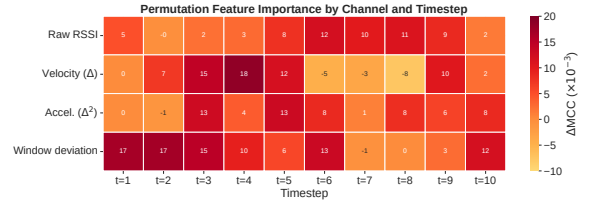


Figure 4. Permutation feature importance by channel and timestep (CNN, TCN, GRU average).

I. Discussion

The complementary profiles of classical and deep methods—KNN leads on clean data (0.922 vs. 0.881), DeepStack on interference (0.859 vs. 0.770)—suggest density-based routing: KNN for sparse single-device conditions where local structure suffices, DeepStack for dense multi-device scenarios where learned feature hierarchies disentangle overlapping signals.

The HPO gain gap between recurrents (+20–23%) and CNN/TCN ($\pm 1\%$) reflects a fundamental architectural difference. CNN kernels operate on fixed-size windows; TCN dilated convolutions naturally match 10-step sequences. Recurrent hidden states must encode all context into a fixed-size vector—at default dim 64, this bottlenecks information flow. HPO discovers larger dimensions.

The augmentation results reinforce this narrative. Models with strong inductive biases extract signal from synthetic diversity (HPO CNN +0.029, HPO TCN +0.018), while recurrents without HPO are confused by it (GRU −0.006). The mechanism: synthetic noise creates spurious local fluctuations that recurrents interpret as temporal dependencies, while CNNs with weight sharing naturally smooth them. Augmentation and HPO are synergistic.

The ensemble comparison reveals why uniform voting fails: All-9 (MCC 0.807) underperforms a single CNN (0.838) because weak classifiers contribute noisy probabilities. Stacking (0.849) partially recovers, but DeepStack (0.859) achieves the best result through per-class Level2Nets that learn class-specific base model importance.

Mamba’s $O(L)$ complexity makes it the natural choice for scaling to longer observation windows (20–30 s), where self-attention’s quadratic cost becomes prohibitive. MetaFusion_GRU (MCC 0.783 on noise vs. 0.758 baseline) demonstrates that context-aware architectures incorporating noise metadata can partially disentangle interference, pointing toward future designs that jointly model signal and environmental context.

Limitations include the controlled environment, fixed 10-second window, MAC randomisation, and untested device heterogeneity. CNN (28K params) suits on-device AP inference; DeepStackEnsemble (1.2M) targets server-side batch inference.

V. CONCLUSION

This paper presented a comprehensive evaluation of deep learning architectures for RSSI-based passenger movement classification in public transport. Building on classical ML baselines (MCC = 0.756), we evaluated 11+ architectures spanning recurrent, convolutional, transformer, state-space, and mixture-of-experts models, supported by class-conditioned data augmentation and exhaustive hyperparameter optimization across four dataset variants.

The two-stage DeepStackEnsemble achieved MCC = 0.859 and accuracy 89.4% on combined data, a 13.6% relative improvement. State-space models (HPO Mamba, up to MCC = 0.850) demonstrated competitive performance with $O(L)$ complexity, establishing them as a viable alternative to transformers for resource-constrained deployment.

Cross-route interference analysis revealed a 9.4% absolute MCC degradation under multi-device noise. Heatmap analysis identified AA as the dominant interferer and AB as the most vulnerable route. Trajectory visualization traced the mechanism: concurrent near-AP transmissions erase AB’s characteristic 20 dBm RSSI descent, the primary discriminative feature for alighting detection. BB proved most noise-resilient because its classification signal—temporal flatness—survives interference-level shifts intact.

Future work should explore attention mechanisms that learn to down-weight interfered timesteps, multi-AP fusion to disambiguate near-AP states through spatial diversity, and integration of movement classification with origin-destination matrix inference for end-to-end passenger flow modeling.

VI. GENERATIVE AI DECLARATION

During the preparation of this manuscript, the authors used DeepSeek as a writing assistant for language improvement, grammar correction, and formatting. All technical content, experimental design, model architectures, results, analysis, and conclusions represent the authors’ original intellectual

contribution. The authors take full responsibility for the accuracy and integrity of the content. No AI-generated figures, fabricated citations, or manipulated results are present in this work.

ACKNOWLEDGMENT

This work is supported by the European Regional Development Fund (FEDER), through the Regional Operational Programme of Centre (CENTRO 2030) of the Portugal 2030 framework [Project inMotion with Nr. 21359 (CENTRO2030-FEDER-02225900)].

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